

3.5. Long- term effects of sucrose on adult cell proliferation in the dentate gyrus

Sucrose consumption during youth affects the hippocampal neurogenesis by reducing the number of new neuronal cells and the neuroproliferation markers DCX and PCNA in the DG [20, 21]. However, it is not yet known whether this effect is sustained over time and contributes to the sucrose-induced memorization deficits observed in the long term. Twenty-four hours after T4 NOR testing, the brains of the sucrose-exposed and control groups were analyzed by immunohistochemistry for PCNA and DCX. PCNA is a marker for neuroprogenitor cell proliferation [56-58], which is expressed during all active phases of the cell cycle and for a short period of time after cells become post-mitotic [59]. DCX is a microtubule-associated protein that is expressed in migrating neuronal precursor cells [60, 61] and is used as a marker for new adult born neurons in the DG [62]. The sucrose-exposed groups had significantly less PCNA or both PCNA-DCX immunopositive cells than the control groups (two-way ANOVA, $F_{\text{Treatment}(1,16)} = 4.734$; $p = 0.0349$ and $F_{\text{Treatment}(1,16)} = 5.769$; $p = 0.0248$, respectively; Figures 8A, C and D). In contrast, no differences were detected between control and sucrose-exposed groups in the number of DCX immunopositive cells in the granular cell layer of the DG (Figures 8B and D). Taken together, these data suggest that continuous sucrose intake leads to a long-term decrease in the number of neuroprogenitor cells and the transition/differentiation of these progenitors to immature neurons, without affecting differentiated neuroblasts. No correlations were detected between PCNA+/ DCX+/ PCNA+-DCX+ expression and the exploration ratios T2-T4 (data not shown).

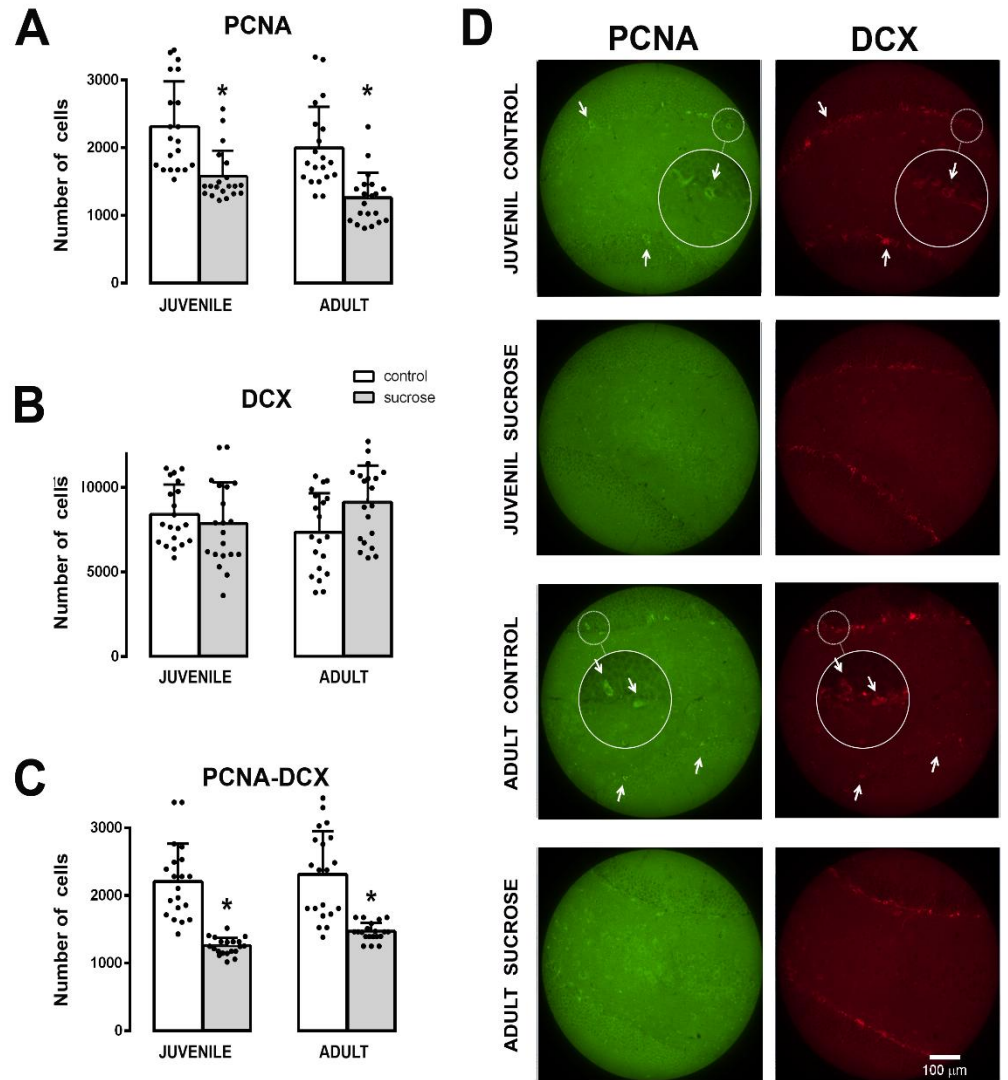


Figure 8. Long-term neurogenic ability in the HIP. Quantification of (A) PCNA-positive cells (PCNA), (B) Doublecortin-positive cells (DCX) and (C) both PCNA-DCX positive cells (merged) per mm² of DG in the dorsal HIP of control rats (white bars) and sucrose-exposed rats that consumed sucrose (gray bars) from juvenile and adult groups. ~~that consumed sucrose during youth or adulthood.~~ Data are presented as mean $\pm$ SD SEM from two independent experiments, N=4, n=5. Tukey's post hoc test, * $p < 0.05$. (D) Representative pictures showing PCNA (first column) and DCX (second column) immunoreactivity in the DG of control and sucrose groups during youth and adulthood. White arrows show both PCNA-DCX immunopositive cells (20X, scale bar=100 μ m).

4. Discussion

In this study we found that early juvenile-adolescent exposure to 10% sucrose in the beverage produces long-term impairment of the object recognition memory, whereas comparable sucrose exposure in adult rats fails to induce such effects. Animals were exposed to three

consecutive trials with different delays after the sample phase (2-h, 24-h and 7-days ITI). For each trial (T2, T3 and T4) a copy of the object presented during the sample phase (T1) and a novel object of different shape, color and height were introduced in the arena. The novel and the familiar objects were counterbalanced and placed in different positions relative to the location they occupied in the previous trial.

When learning performances of control and sucrose-exposed groups were compared differences by the age of sucrose exposure were noticed. Two hours, 24h or 7 days after the sample phase T1, adult animals from juvenile sucrose-exposed group ~~with a history of sucrose consumption during youth~~ failed to distinguish between the familiar and the novel objects. This suggests that early exposure to unlimited sucrose could affect the initial encoding and acquisition of the object, since the same familiar object was presented along T2-T4 trials. Nonetheless, the difficulty in differentiating the novel object from the familiar one along T2-T4 trials could also be the expression of impaired memory consolidation or memory retrieval. More studies are required to distinguish between these possibilities, which are not mutually exclusive. Notably, since the short duration between the familiarization T1 and the test T2 trials is primarily dependent on the perirhinal cortex, these data show that early life sugar consumption impairs perirhinal cortex-dependent memory besides hippocampal-dependent memory in the adulthood [9, 32].

No differences were found between the sucrose-exposed group and its age control in the total time spent exploring the objects neither during the sample phase T1 nor during T2-T4 trials, indicating that the alterations found in the learning performance of sucrose-exposed animals from juvenile group ~~during youth~~ is not due to an attention deficit but likely results from impaired NOR. Furthermore, the sucrose-exposed animals exhibited identical locomotor activity levels as shown by the similar number of line crossings during the test trials relative to their corresponding age control. Notably, we have previously reported anxiety-like behavior measured by a reduced exploration of the center of the open field [7], however in this study animals were habituated for 5 consecutive days before tested and the arena dimensions were smaller than the one used in the open field. Altogether these observations show that animals under the NOR testing conditions do not show anxiety and, consequently, the cognitive defects observed result of learning and memory impairment and not from the interference of stress during the study.

Our results are in accordance with those reporting sugar-induced impairments in NOR memory task [22, 28, 63]. However, most of the studies looked at the immediate effect of sugar on memory, but not whether the effect persists over time. Here we show long-term sugar-induced disturbances only in animals that had unlimited access to sucrose during youth. In contrary, some studies have shown that perirhinal-dependent memory was not disrupted following either intermittent [23, 24] or continuous sucrose consumption in rats [8, 30]. However, these latest studies [8, 30] were carried out in adult rats and like us, no effects were observed in this group of age. Even more, a

recent study in young animals have shown that intermittent access to a 10% sucrose solution for 2 weeks can disrupt perirhinal-dependent memory when objects have increased similarity and share multiple features [29]. In this study female rats consumed significantly more sucrose solution than males performing worst in the memory tasks, and the authors finally suggest that with a prolonged consumption of sucrose (more than 2 weeks) it could also further decrease the NOR performance in males. Altogether these data indicate that ~~young~~ juvenile animals are more susceptible to the adverse effects of sucrose consumption on cognition. It should be noted that these cognitive changes are observed in the absence of weight differences [7, 20, 21] suggesting that the metabolic disturbances by the diet, rather than obesity, underpins the cognitive deficits.

Prolonged consumption of high-fat high-sugar diets has deleterious effects on hippocampal neurogenesis [76, 77]. In juvenile ~~young~~ rats, intermittent access to sucrose decreases the neuroproliferation markers DC and PCNA [21], whereas unlimited access to sucrose decreases the hippocampal neurogenesis in a long-term study measured by BrdU [20]. However, the link between the long-term deficits in memory recognition and the changes in hippocampal neurogenesis from sucrose consumption has yet to be explored. In our study, a reduction of neuroprogenitors cells and their maturation to immature neurons was observed by sucrose exposure regardless of the age at which this exposure occurs. No changes within differentiated migrating immature neurons were found suggesting that sucrose consumption principally affects the earlier stages of neurogenesis. In accordance, a recent study in mice shows the same effect after 12 weeks of continuous sucrose treatment [78]. Altogether these results indicate that the decreased proliferation and differentiation of newborn neurons within the DG induced by sucrose consumption is maintained even after 25 days of sugar removal. Despite the reductions in neuroproliferation observed in the adult sucrose-exposed group, performance of the NOR task was not affected in this group of age and no correlation between the performances of the memory test (T2-T4) and the expression of PCNA+, DCX+ or both PCNA+-DCX+ were found, indicating that the deficits in the NOR are not associated with changes in hippocampal neurogenesis. In accordance, previous research has shown that a reduced neuroproliferation in the DG by caffeine-sucrose treatment does not affect place and long-term object memory performance [24]. Indeed, adult neurogenesis in the DG is proposed to be a neural substrate for specific aspects of memory formation, in particular “pattern separation”, a process distinguishing and coding similar patterns of neural activity as distinct representations during memory encoding and storage [79]. Then, adult neurogenesis provides an added functionality to the HIP when the cognitive load is taxed [80]. Supporting our findings, 10% sucrose consumption reduces neurogenesis together with an impaired performance in the Spontaneous Location Recognition (SLR) task, only when the spatial arrangement of the identical objects are closer together (s-SLR; high

pattern separation), but not when the objects are further apart (d-SLR; low pattern separation) [21].